
Week 34: Probability theory

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These notes are written by myself, and errors may and will occur. When in doubt, trust the book and Gunnars lectures!

1. Theory based on Karr (1993)

Informally, a *random experiment* has the following objects:

1. Sample space: Set Ω with possible outcomes $\omega \in \Omega$.
2. Events: Subsets of Ω for which probabilities are defined.
3. Measurable space: Sample space Ω equipped with a family of events $\mathcal{E}$.

Definition 1.1. (*σ -algebra*) A σ -algebra $\mathcal{E}$ on Ω is a family of subsets of Ω satisfying

- $\Omega \in \mathcal{E}$
- If $A \in \mathcal{E}$, then $A^C = \Omega \setminus A \in \mathcal{E}$.
- If $A_1, A_2, \dots \in \mathcal{E}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{E}$.

Definition 1.2. (*Events*) The family of events $\mathcal{E}$ is a σ -algebra on Ω .

Definition 1.3. (*Probability*) A probability on $(\Omega, \mathcal{E})$ is a function $P : \mathcal{E} \rightarrow \mathbb{R}$ such that

- $P(A) \geq 0$ for all $A \in \mathcal{E}$.
- $P(\Omega) = 1$.
- If $A_1, A_2, \dots \in \mathcal{E}$ are pairwise disjoint, then $P(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$.

We often call the tuple $(\Omega, \mathcal{E}, P)$ a *probability space*.

Definition 1.4. (*Random variable*) A random variable (RV) is a function $X : \Omega \rightarrow \mathbb{R}$ such that

$$X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{E}$$

for every $B \in \mathcal{B}(\mathbb{R})$. Here $\mathcal{B}(\mathbb{R})$ denotes the Borel σ -algebra on $\mathbb{R}$, which is the smallest σ -algebra containing all open intervals in $\mathbb{R}$ (Karr, 1993).

Proposition 1.5. (*Probability associated with RV*) If X is a RV, then the function $P_X : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ defined by

$$P_X(B) := P(X \in B) = P(X^{-1}(B))$$

is a probability on $\mathbb{R}$. We will prove this in the last problem.

Definition 1.6. (*Cumulative distribution function associated with an RV*) The cumulative distribution function (CDF) of X is the function $F_X : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_X(x) := P_X((-\infty, x]) = P(X \leq x).$$

Definition 1.7. (*Simple RV*) A RV X is simple if it takes on only finitely many values. We can write this as $X(\Omega) = \{X(\omega) \mid \omega \in \Omega\} = \{x_1, \dots, x_n\}$ for a finite n . Defining $A_i = \{\omega \in \Omega : X(\omega) = x_i\}$, we have that $A_1, \dots, A_n$ is a disjoint partition of Ω , and X can be expressed as $X = \sum_{i=1}^n x_i \mathbb{1}_{A_i}$.

Definition 1.8. (*Expectation of simple RVs*) The expectation of a simple RV $X = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is defined as $\mathbb{E}[X] = \sum_{i=1}^n a_i P(A_i)$.

2. Problems

Problem 2.1. (Sample spaces, events and measurable spaces) Let $\Omega = \{1, 2, 3\}$ and $\mathcal{E} = \{\emptyset, \Omega, \{1\}, \{2, 3\}\}$.

1. Show that $(\Omega, \mathcal{E})$ is a measurable space.
2. Is $\{1, 2\}$ an event?
3. How many subsets of Ω are events?
4. Every event is a set, but every set is not an event. Explain this.

Solution.

1. Need to show that $\mathcal{E}$ is a σ -algebra on Ω . All the conditions listed in [Definition 1.1](#) are easy to verify.
2. As $\{1, 2\} \notin \mathcal{E}$, it is not an event.
3. There are four events, as listed in $\mathcal{E}$.
4. Only the sets in $\mathcal{E}$ are events. There are a total of $2^3 = 8$ subsets of Ω , but only four of them are events.

Problem 2.2. (*Probabilities and RVs*) Let $X : \Omega \rightarrow \mathbb{R}$ be such that $(X \leq q)$ is an event for all $q \in \mathbb{Q}$.

1. Prove that $X^{-1}((-\infty, x]) = (X \leq x)$ is an event for every $x \in \mathbb{R}$.
2. Prove that $X^{-1}(I)$ is an event for every interval $I \subseteq \mathbb{R}$.
3. Define

$$\mathcal{E}_X := \{B \subseteq \mathbb{R} : X^{-1}(B) \in \mathcal{E}\},$$

and that $\mathcal{E}_X$ is a σ -algebra on $\mathbb{R}$.

4. Use the preceding parts to prove that X is a RV. Conclude that its CDF $F_X(x)$ is defined for every $x \in \mathbb{R}$.

Solution.

1. For each positive integer n , choose $q_n \in \mathbb{Q}$ such that

$$x < q_n < x + \frac{1}{n}.$$

Such rational numbers exist because there is a rational number between any two different real numbers.

If $X \leq x$, then $X \leq q_n$ for every n , and if $X > x$, then eventually $q_n < X$. Therefore we have

$$(X \leq x) = \bigcap_{n=1}^{\infty} (X \leq q_n). \quad (1)$$

Each $(X \leq q_n)$ is an event by assumption. A σ -algebra is closed under countable intersections, since

$$\bigcap_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} E_n^C \right)^C.$$

Thus $(X \leq x)$ is an event.

2. We also have that strict inequalities define events. This is because

$$(X < x) = \bigcup_{q \in \mathbb{Q}, q < x} (X \leq q).$$

which is a countable union of events. Complements give

$$(X > a) = (X \leq a)^C, \quad (X \geq a) = (X < a)^C.$$

Any interval can be made by intersecting sets of these four types. For example,

$$(a < X \leq b) = (X > a) \cap (X \leq b),$$

meaning $X^{-1}(I) = (X \in I)$ is an event whenever I is an interval.

3. First, $\mathbb{R} \in \mathcal{E}_X$ because $X^{-1}(\mathbb{R}) = \Omega \in \mathcal{E}$. If $B \in \mathcal{E}_X$, then

$$X^{-1}(B^C) = (X^{-1}(B))^C \in \mathcal{E},$$

so $B^C \in \mathcal{E}_X$. Finally, if $B_1, B_2, \dots \in \mathcal{E}_X$ are pairwise disjoint, then

$$X^{-1}\left(\bigcup_{n=1}^{\infty} B_n\right) = \bigcup_{n=1}^{\infty} X^{-1}(B_n) \in \mathcal{E}.$$

Therefore $\bigcup_{n=1}^{\infty} B_n \in \mathcal{E}_X$, and $\mathcal{E}_X$ is a σ -algebra on $\mathbb{R}$.

4. Let $\mathcal{I}$ be the family of open intervals in $\mathbb{R}$. Part 2 shows that $\mathcal{I} \subseteq \mathcal{E}_X$. Since the Borel σ -algebra $\mathcal{B}(\mathbb{R})$ is the smallest σ -algebra containing all open intervals, we have

$$\mathcal{B}(\mathbb{R}) \subseteq \mathcal{E}_X.$$

Consequently, $X^{-1}(B) \in \mathcal{E}$ for every $B \in \mathcal{B}(\mathbb{R})$, which proves that X is a RV. Furthermore, $(X \leq x)$ is an event for every $x \in \mathbb{R}$, so

$$F_X(x) = P(X \leq x)$$

is defined for every $x \in \mathbb{R}$.

Problem 2.3. (Proving some important properties of expectation for simple RVs) Let X and Y be simple RVs, and let $\alpha, \beta \in \mathbb{R}$. Prove

- Constants are preserved: If $X = c$ for some $c \in \mathbb{R}$, then $\mathbb{E}[X] = c$.
- Monotonicity: If $X \leq Y$, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$.
- Linearity: $\mathbb{E}[\alpha X + \beta Y] = \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]$.
- Relation to probability: If $A \in \mathcal{E}$, then $\mathbb{E}[\mathbb{1}(A)] = P(A)$.

Solution. Denote the simple RVs as $X = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ and $Y = \sum_{j=1}^m b_j \mathbb{1}_{B_j}$.

- If $X(\omega) = c \in \mathbb{R}$ for every $\omega \in \Omega$, then $\mathbb{E}[X] = \mathbb{E}[c] = cP(\Omega) = c$.
- If $X \leq Y$, then $a_i \leq b_j$ whenever $A_i \cap B_j \neq \emptyset$. Since both A_i and B_j give a disjoint partitions of Ω , we have

$$\mathbb{E}[X] = \sum_{i=1}^n a_i P(A_i) = \sum_{i=1}^n \sum_{j=1}^m a_i P(A_i \cap B_j) \leq \sum_{i=1}^n \sum_{j=1}^m b_j P(A_i \cap B_j) = \sum_{j=1}^m b_j P(B_j) = \mathbb{E}[Y].$$

- We can express

$$\alpha X + \beta Y = \sum_{i=1}^n \sum_{j=1}^m (\alpha a_i + \beta b_j) \mathbb{1}_{A_i \cap B_j},$$

meaning that $\alpha X + \beta Y$ is also simple. Thus

$$\begin{aligned} \mathbb{E}[\alpha X + \beta Y] &= \sum_{i=1}^n \sum_{j=1}^m (\alpha a_i + \beta b_j) P(A_i \cap B_j) \\ &= \alpha \sum_{i=1}^n \sum_{j=1}^m a_i P(A_i \cap B_j) + \beta \sum_{i=1}^n \sum_{j=1}^m b_j P(A_i \cap B_j) \\ &= \alpha \sum_{i=1}^n a_i P(A_i) + \beta \sum_{j=1}^m b_j P(B_j) \\ &= \alpha \mathbb{E}[X] + \beta \mathbb{E}[Y]. \end{aligned}$$

- As $\mathbb{1}_A = 1 \cdot \mathbb{1}_A + 0 \cdot \mathbb{1}_{A^C}$, the indicator function is clearly simple. This means

$$\mathbb{E}[\mathbb{1}(A)] = 1 \cdot P(A) + 0 \cdot P(A^C) = P(A).$$

Problem 2.4. (Towards a more general definition of the expectation) Let $X : \Omega \rightarrow \mathbb{R}$ be an RV, and let $\varphi : \mathcal{B}(\mathbb{R}) \rightarrow \mathbb{R}$ be a simple and measurable function.

1. Show that $P_X(B) = P(X \in B)$ defines a probability on $(\mathbb{R}, \mathcal{E}_X)$.
2. Show that $\mathbb{E}[\varphi(X)] = \mathbb{E}_X[\varphi]$ with $\mathbb{E}_X[\cdot]$ denoting the expectation with respect to the probability P_X .

Solution.

1. For P_X to define a probability distribution on $(\mathbb{R}, \mathcal{E}_X)$, we simply check the three conditions:

- With $A \in \mathcal{E}_X$ we have $P_X(B) = P(X \in B) \geq 0$ since $P(\cdot)$ is a probability.
- $P_X(\mathbb{R}) = P(X \in \mathbb{R}) = P(\Omega) = 1$.
- With pairwise disjoint $B_1, B_2, \dots \in \mathcal{E}_X$, we have

$$P_X\left(\bigcup_{i=1}^{\infty} B_i\right) = P\left(X^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right)\right) = P\left(\bigcup_{i=1}^{\infty} X^{-1}(B_i)\right) = \sum_{i=1}^{\infty} P(X^{-1}(B_i)) = \sum_{i=1}^{\infty} P_X(B_i).$$

2. As φ simple we can write $\varphi = \sum_{i=1}^n a_i \mathbb{1}_{B_i}$ for some $a_i \in \mathbb{R}$ and $B_i \in \mathcal{E}_X$. Then, by linearity of the expectation, we see

$$\mathbb{E}[\varphi(X)] = \mathbb{E}\left[\sum_{i=1}^n a_i \mathbb{1}_{B_i}(X)\right] = \sum_{i=1}^n a_i \mathbb{E}[\mathbb{1}_{B_i}(X)] = \sum_{i=1}^n a_i P(X \in B_i) = \sum_{i=1}^n a_i P_X(B_i) = \mathbb{E}_X[\varphi].$$

Proposition 2.1. (General definition of the expectation) A more general definition of expectation is using the Lebesgue integral

$$\mathbb{E}[\varphi(X)] = \int_{\Omega} \varphi(X) dP$$

which is beyond the scope of this course. However, for a RV $X : \Omega \rightarrow \mathbb{R}$ with $X(\omega) = x$, we can rewrite this as an integral over $\mathbb{R}$

$$\mathbb{E}[\varphi(X)] = \int_{\Omega} \varphi(X(\omega)) dP(\omega) = \int_{\mathbb{R}} \varphi(x) dP(X^{-1}(x)) = \int_{\mathbb{R}} \varphi(x) dP_X(x)$$

For a simple RV, the $dP_X(x)$ is a discrete measure, and the integral reduces to the sum in [Definition 1.8](#). If X is a continuous RV, then the $dP_X(x) = f_X(x) dx$, where f_X is the probability density function of X . This gives the familiar expression

$$\mathbb{E}[\varphi(X)] = \int_{\mathbb{R}} \varphi(x) f_X(x) dx,$$

that you might remember from the first course in statistics.

References

Karr, A. F. *Probability*. Springer-Verlag, 1993. <https://doi.org/10.1007/978-1-4612-0891-4>